GATE CS — Programming & Data Structures

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
C programming: pointers, recursion, output prediction	58	93	2
Trees, BST, heaps	29	44	2
Stacks, queues, linked lists	16	24	2
Total	**103**	**161**	**6**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

C programming: pointers, recursion, output prediction

Output of C programs: pointers, static variables, recursion, arrays/strings. · 58 questions, 93 marks over 15 papers (≈6.2 marks/paper)

A. Past-paper questions (58)

2026 CS1 · Q.27 · 1 mark

Consider the following C statements: char *str1 = "Hello; /* Statement S1 */ char *str2 = "Hello;"; /* Statement S2 */ int *str3 = "Hello"; /* Statement S3 */ Which of the following options is/are correct?
(A) S1 and S2 have syntactic errors
(B) S2 has a lexical error and S3 has a syntactic error
(C) S1 has a lexical error and S3 has a semantic error
(D) S1 has a syntactic error and S3 has a semantic error

2026 CS1 · Q.34 · 1 mark

Consider the following program in C: #include <stdio.h> void func(int i, int j) { if(i < j) { int i = 0; while (i < 10) { j += 2; i++; } } printf("%d", i); } int main() { int i = 9, j = 10; func(i, j); return 0; } The output of the program is _____. (answer in integer) Note: Assume that the program compiles and runs successfully.

2026 CS1 · Q.61 · 2 marks

Consider the recursive functions represented by the following code segment: int bar(int n) { if (n == 1) return 0; else return 1 + bar(n/2); } int foo(int n){ if (n == 1) return 1; else return 1 + foo(bar(n)); } The smallest positive integer n for which foo(n) returns 5 is _____. (answer in integer) Note: Ignore syntax errors (if any) in the function.

2026 CS2 · Q.19 · 1 mark

Consider the following three ANSI-C programs, P1, P2, and P3. P1 P2 P3 #include <stdio.h> #include <stdio.h> #include <stdio.h> int a=5; int main(){ int main(){ int main(){ int a=5; int a=5; int a=7; int a=7; float a=7; return(0); return(0); return(0); } } } Which one of the following statements is true?
(A) Only P1 will compile without any error
(B) Only P2 will compile without any error
(C) Only P3 will compile without any error
(D) All three programs P1, P2, and P3 will compile without any error

2026 CS2 · Q.60 · 2 marks

Consider the following ANSI-C program. #include <stdio.h> int main(){ int *ptr, a, b, c; a=5; b=11; c=20; ptr=&a; *ptr=c; ptr=&c; a=*(&b); c=*ptr-a; printf("%d",c); return(0); } The output of this program is _____. (answer in integer) Note: Assume that the program compiles and runs successfully.

2026 CS2 · Q.61 · 2 marks
Consider the following ANSI-C function. int func(int start, int end){ int length=end+1-start; if((length<1)||(start<0)||(end<0)){ return(0); } if(length%3==0){ return(func(start+1, end)); } else if(length%3==1){ return(1+func(start, end-1)); } else { return(func(start+2, end)); } } The maximum possible value that can be returned from this function is _____. (answer in integer) Note: Ignore syntax errors (if any) in the function.

2025 CS1 · Q.34 · 1 mark
#include <stdio.h> void foo(int *p, int x){ *p=x; } int main(){ int *z; int a = 20, b = 25; z = &a; foo(z,b); printf("%d",a); return 0; } The output of the given C program is _____. (Answer in integer)

2025 CS1 · Q.61 · 2 marks
#include <stdio.h> int foo(int S[],int size){ if(size == 0) return 0; if(size == 1) return 1; if(S[0] != S[1]) return 1+foo(S+1,size-1); return foo(S+1,size-1); } int main(){ int A[]= {0,1,2,2,2,0,0,1,1}; printf("%d",foo(A,9)); return 0; } The value printed by the given C program is _____ . (Answer in integer)

2025 CS1 · Q.63 · 2 marks
Consider the following C program: #include <stdio.h> int gate (int n) { int d, t, newnum, turn; newnum = turn = 0; t=1; while (n>=t) t *= 10; t /=10; while (t>0) { d = n/t; n = n%t; t /= 10; if (turn) newnum = 10*newnum + d; turn = (turn + 1) % 2; } return newnum; } int main () { printf ("%d", gate(14362)); return 0; } The value printed by the given C program is _____ . (Answer in integer)

2025 CS2 · Q.19 · 1 mark
Consider the following C program: #include <stdio.h> void stringcopy(char *, char *); int main(){ char a[30] = "@#Hello World!"; stringcopy(a, a + 2); printf("%s\n", a); return 0; } void stringcopy(char *s, char *t) { while(*t) *s++ = *t++; } Which ONE of the following will be the output of the program?
(A) @#Hello World!
(B) Hello World!
(C) ello World!
(D) Hello World!d!

2025 CS2 · Q.33 · 1 mark
int x=126,y=105; do { if(x>y) x=x-y; else y=y-x; } while(x!=y); printf("%d",x); The output of the given C code segment is _____. (Answer in integer)

2025 CS2 · Q.62 · 2 marks
Consider the following C program: #include <stdio.h> int main(){ int a; int arr[5] = {30,50,10}; int *ptr; ptr = &arr[0] + 1; a = *ptr; (*ptr)++; ptr++; printf("%d", a + (*ptr) + arr[1]); return 0; } The output of the above program is _____. (Answer in integer)

2025 CS2 · Q.63 · 2 marks
Consider the following C program: #include <stdio.h> int g(int n) { return (n+10); } int f(int n) { return g(n*2); } int main() { int sum, n; sum=0; for (n=1; n<3; n++) sum += g(f(n)); printf ("%d", sum); return 0; } The output of the given C program is _____. (Answer in integer)

2024 CS1 · Q.18 · 1 mark

Consider the following C program: #include <stdio.h> int main(){ int a = 6; int b = 0; while(a < 10) { a = a / 12 + 1; a += b;} printf(”%d”, a); return 0;} Which one of the following statements is CORRECT?

(A) The program prints 9 as output

(B) The program prints 10 as output

(C) The program gets stuck in an infinite loop

(D) The program prints 6 as output

2024 CS1 · Q.19 · 1 mark

Consider the following C program: #include <stdio.h> void fX(){ void fX(); char a; int main(){ if((a=getchar()) != ’\n’) fX(); fX(); return 0;} if(a != ’\n’) putchar(a);} Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

(A) The program will not terminate

(B) The program will terminate with no output

(C) The program will terminate with 4321 as output

(D) The program will terminate with 1234 as output

2024 CS1 · Q.48 · 2 marks

Consider the following C function definition. int f(int x, int y) { for (int i=0; i<y; i++) { x=x+x+y; } return x; } Which of the following statements is/are TRUE about the above function?

(A) If the inputs are x=20, y=10, then the return value is greater than 220

(B) If the inputs are x=20, y=20, then the return value is greater than 220

(C) If the inputs are x=20, y=10, then the return value is less than 210

(D) If the inputs are x=10, y=20, then the return value is greater than 220

2024 CS2 · Q.13 · 1 mark

Consider the following C program. Assume parameters to a function are evaluated from right to left. #include <stdio.h> int g(int p) { printf("%d", p); return p; } int h(int q) { printf("%d", q); return q; } void f(int x, int y) { g(x); h(y); } int main() { f(g(10),h(20)); } Which one of the following options is the CORRECT output of the above C program?

(A) 20101020

(B) 10202010

(C) 20102010

(D) 10201020

2024 CS2 · Q.33 · 1 mark

Consider the following C function definition. int fX(char *a){ char *b = a; while(*b) b++; return b - a;} Which of the following statements is/are TRUE?

(A) The function call fX(”abcd”) will always return a value Assuming a character array c is declared as char c[] = ”abcd” in main(),

(B) the function call fX(c)will always return a value

(C) The code of the function will not compile

(D) Assuming a character pointer c is declared as char *c = ”abcd” in main(), the function call fX(c)will always return a value

2024 CS2 · Q.36 · 2 marks

What is the output of the following C program? #include <stdio.h> int main() { double a[2]={20.0, 25.0}, *p, *q; p = a; q = p + 1; printf(”%d,%d”, (int)(q – p), (int)(*q – *p)); return 0;}

(A) 4,8

(B) 1,5

(C) 8,5

(D) 1,8

2024 CS2 · Q.42 · 2 marks

be a sequence of array locations with consecutive indices. The C code snippet given below has been written to compute the length of the longest subarray of X that contains at most two distinct integers. The code has two missing expressions labelled (P)and (Q). int first=0, second=0, len1=0, len2=0, maxlen=0; for (int i=0; i < n; i++) { if (X[i] == first) { len2++; len1++; } else if (X[i] == second) { len2++; len1 = (P); second = first; } else { len2 = (Q); len1 = 1; second = first; } if (len2 > maxlen) { maxlen = len2; } first = X[i]; } Which one of the following options gives the CORRECT missing expressions? (Hint: At the end of the i-th iteration, the value of len1 is the length of the longest subarray ending with X[i] that contains all equal values, and len2 is the length of the longest subarray ending with X[i] that contains at most two distinct values.)

(A) (P) len1+1 (Q) len2+1

(B) (P) 1 (Q) len1+1

(C) (P) 1 (Q) len2+1

(D) (P) len2+1 (Q) len1+1

2023 · Q.35 · 1 mark

The integer value printed by the ANSI-C program given below is . #include<stdio.h> int funcp(){ static int x = 1; x++; return x; } int main(){ int x,y; x = funcp(); y = funcp()+x; printf("%d\n", (x+y)); return 0; } CS

2023 · Q.36 · 2 marks

Consider the following program: int main() int f1() int f2(int X) int f3() { { { { f1(); return(1); f3(); return(5); f2(2); } if (X==1) } f3(); return f1(); return(0); else } return (X*f2(X-1)); } Which one of the following options represents the activation tree corresponding to the main function? main f1 f2 f3 f3 f2

(A) f3 f1 main f1 f2 f3

(B) f3 f1 main f1 f2

(C) f3 f1 main f1 f2 f3

(D) f3 f2 f1 CS

2022 · Q.21 · 1 mark

What is printed by the following ANSI C program? #include<stdio.h> int main(int argc, char *argv[]) { int x = 1, z[2] = {10, 11}; int *p = NULL; p = &x; *p = 10; p = &z[1]; *(&z[0] + 1) += 3; printf("%d, %d, %d\n", x, z[0], z[1]); return 0; }

(A) 1, 10, 11

(B) 1, 10, 14

(C) 10, 14, 11

(D) 10, 10, 14

2022 · Q.43 · 2 marks
What is printed by the following ANSI C program? #include<stdio.h> int main(int argc, char *argv[]) { int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9}, {10, 11, 12, 13, 14, 15, 16, 17, 18}, {19, 20, 21, 22, 23, 24, 25, 26, 27}}; int i = 0, j = 0, k = 0; for(i = 0; i < 3; i++){ for(k = 0; k < 3; k++) printf("%d ", a[i][j][k]); printf("\n"); } return 0; }
(A) 1 2 3 10 11 12 19 20 21
(B) 1 4 7 10 13 16 19 22 25
(C) 1 2 3 4 5 6 7 8 9
(D) 1 2 3 13 14 15 25 26 27

2022 · Q.44 · 2 marks
What is printed by the following ANSI C program? #include<stdio.h> int main(int argc, char *argv[]){ char a = 'P'; char b = 'x'; char c = (a & b) + '*'; char d = (a | b) - '-'; char e = (a ^ b) + '+'; printf("%c %c %c\n", c, d, e); return 0; } ASCII encoding for relevant characters is given below
(A) z K S
(B) 122 75 83
(C) * - +
(D) P x +

2021 · Q.37 · 2 marks [OCR]
Consider the following ANSI C program. #include <stdio.h> int main() { int i, j, count; count = 0; i=0; for (j = -3; j <= 3; j++) { if ((j >= 0) && (it+)) count = count + j; } count = count + i; printf("%d", count); return 0; } Which one of the following options is correct?
(A) | The program will not compile successfully.
(B) | The program will compile successfully and output 10 when executed.
(C) | The program will compile successfully and output 8 when executed.
(D) | The program will compile successfully and output 13 when executed. CS, Set-1 - Copyright © GATE 2021

2021 · Q.48 · 2 marks [OCR]
Consider the following ANSI C function: int SimpleFunction(int Y[], int n, int x) 5! int total = Y[0], loopIndex; for (loopIndex = 1; loopIndex <= n - 1; loopIndex++) total = x * total + Y[loopIndex] ; return total; } Let Z be an array of 10 elements with Z[i]=1, for all i such that 0 < i < 9. The value returned by SimpleFunction(Z, 10, 2) is

2020 · Q.22 · 1 mark [OCR]
Consider the following C program. #include <stdio.h> int main() { int a[4][5]={{1, 2, 3, 4, 5}, {6, 7, 8, 9, 10}, {ll, 12, 23; 14, 15, (16, 17, 18 , 19, 20}}; printf ("Sd\n", *(*(at**a+2)+3)); return (0); The output of the program is

2020 · Q.46 · 2 marks [OCR]
Consider the following C functions. int funl(int n) { int fun2(int n) { static int i = 0; static int i = 0; af (n > 0) £ if (n> 0) { ++i; i =i + funl(n); funl(n-1); fun2 (n-1); 7 } return (i); return (i); The return value of fun2 (5) is

2020 · Q.48 · 2 marks [OCR]
Consider the following C functions. int tob(int b, int* arr) { int pp(int a,int b) { int i; int arr[20]; for(i=0; b>O; i++) { int i, tot = 1, ex, len; if(b%2) arr[i]=1; ex = a; else arr[i]=0; len = tob(b,arr); b = b/2; for(i=0; i<len; i++) { } if (arr[i]==1) return (i); tot = tot * ex; ex = ex * ex; } return (tot); The value returned by pp (3,4) is

2019 · Q.18 · 1 mark [OCR]
Consider the following C program: #include <stdio.h> int jumble(int x, int y){ x=2*xty; return x; int main(){ int x=2, y=5; y=jumble(y,x); x= jumble(y,x); printf£(""%d \n", x); return 0; } The value printed by the program is

2019 · Q.24 · 1 mark [OCR]
Consider the following C program: #include <stdio.h> int main(){ int arr[]= {1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4; printf("%d\n", ip[1]); return 0; } The number that will be displayed on execution of the program is. cs S/S GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.26 · 2 marks [OCR]
Consider the following C function. void convert(int n){ if(n<0) print£("%d",n); else { convert(n/2); printf("%d",n%2); } Which one of the following will happen when the function convert is called with any positive integer n as argument?
(A) It will print the binary representation of n and terminate
(B) It will print the binary representation of n in the reverse order and terminate
(C) It will print the binary representation of n but will not terminate
(D) It will not print anything and will not terminate

2019 · Q.27 · 2 marks [OCR]
Consider the following C program: #include <stdio.h> int r(){ static int num=7; return num--; int main(){ for (r()r()ix()) printf("%d",r()); return 0; Which one of the following values will be displayed on execution of the programs?
(A) 41
(B) 52
(C) 63
(D) 630 cs 6/15 GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.52 · 2 marks [OCR]
Consider the following C program: #include <stdio.h> int main(){ float sum = 0.0, j = 1.0, i = 2.0; while (i/j > 0.0625){ jsit+ii sum = sum + i/j; printf("%f\n", sum); } return 0; } The number of times the variable sum will be printed, when the above program is executed, is cs 12/15 GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.53 · 2 marks [OCR]
Consider the following C program: #include <stdio.h> int main() { int a[] = {2, 4, 6, 8, 10}; int i, sum = 0, *b=a + 4; for (i = 0; i < 5; i++) sum = sum + (*b — i) — *(b— i); printf ("%d\n", sum); return 0; } The output of the above C program is

2018 · Q.2 · 1 mark

Consider the following C program. #include<stdio.h> struct Ournode{ char x,y,z; }; int main(){ struct Ournode p = {'1', '0', 'a'+2}; struct Ournode *q = &p; printf ("%c, %c", *((char*)q+1), *((char*)q+2)); return 0; } The output of this program is:

(A) 0, c

(B) 0, a+2

(C) '0', 'a+2'

(D) '0', 'c'

2018 · Q.21 · 1 mark

Consider the following C program: #include <stdio.h> int counter = 0; int calc (int a, int b) { int c; counter++; if (b==3) return (a*a*a); else { c = calc(a, b/3); return (c*c*c); } } int main (){ calc(4, 81); printf ("%d", counter); } The output of this program is _____.

2018 · Q.29 · 2 marks

Consider the following C program: #include<stdio.h> void fun1(char *s1, char *s2){ char *tmp; tmp = s1; s1 = s2; s2 = tmp; } void fun2(char **s1, char **s2){ char *tmp; tmp = *s1; *s1 = *s2; *s2 = tmp; } int main(){ char *str1 = "Hi", *str2 = "Bye"; fun1(str1, str2); printf("%s %s ", str1, str2); fun2(&str1, &str2); printf("%s %s", str1, str2); return 0; } The output of the program above is

(A) Hi Bye Bye Hi

(B) Hi Bye Hi Bye

(C) Bye Hi Hi Bye

(D) Bye Hi Bye Hi

2018 · Q.32 · 2 marks

Consider the following C code. Assume that unsigned long int type length is 64 bits. unsigned long int fun(unsigned long int n){ unsigned long int i, j = 0, sum = 0; for (i = n; i > 1; i = i/2) j++; for (; j > 1; j = j/2) sum++; return(sum); } The value returned when we call fun with the input 240 is

(A) 4

(B) 5

(C) 6

(D) 40

2018 · Q.45 · 2 marks

Consider the following program written in pseudo-code. Assume that x and y are integers. Count(x,y) { if (y != 1){ if (x != 1) { print("*"); Count(x/2, y); } else { y = y-1; Count(1024, y); } } } The number of times that the print statement is executed by the call Count(1024,1024) is _____.

2017 CS1 · Q.11 · 1 mark [OCR]
Consider the C struct defined below: struct data { int marks [100]; char grade; int cnumber; he struct data student; The base address of student is available in register Rl. The field student.grade can be accessed efficiently using
(A) | Post-increment addressing mode, (R1)+
(B) Pre-decrement addressing mode, —(R1)
(C) Register direct addressing mode, R1
(D) Index addressing mode. X(R1). where X is an offset represented in 2s complement 16-bit representation. =====PAGE 7=====

2017 CS1 · Q.13 · 1 mark [OCR]
Consider the following C code: #include <stdio.h> int *assignval(int *x, int val) { *x = val; return x; } void main () { int *x = malloc(sizeof (int)); if (NULL == x) return; xX = assignval(x,0); if(x) { x = (int *)malloc(sizeof(int)); if (NULL == x) return; x = assignval(x,10)? } PLrinel (“sd\n""’,, +X)? free (x); } The code suffers from which one of the following problems:
(A) compiler error as the return of malloc is not typecast appropriately
(B) compiler error because the comparison should be made as x == NULL and not as shown
(C) compiles successfully but execution may result in dangling pointer
(D) compiles successfully but execution may result in memory leak

2017 CS1 · Q.35 · 2 marks [OCR]
Consider the following two functions. void funl(int n) { void fun2(int n) { if(n == 0) return; if(n == 0) return; printf ("%d", n)? PEINnEL ("SdTM; nN) ? fun2(n —- 2); funl (++n); print£ ("$d".,. 1)% princr("SdTM,, wi) iz } } The output printed when fun! (5) is called is
(A) 53423122233445
(B) 53423120112233
(C) 53423122132435
(D) 53423120213243

2017 CS1 · Q.36 · 2 marks [OCR]
Consider the C functions foo and bar given below: int foo(int val) { int x = 0; while(val > 0) x = x + foo(val--); } return val; } int bar(int wal) { int x = 0; while(val > 0) { x = x + bar(val-1); } return val; } Invocations of foo (3) and bar (3) will result in:
(A) _ Return of 6 and 6 respectively.
(B) Infinite loop and abnormal termination respectively.
(C) — Abnormal termination and infinite loop respectively.
(D) _ Both terminating abnormally. =====PAGE 17=====

2017 CS1 · Q.53 · 2 marks [OCR]
Consider the following C program. #include <stdio.h> #include <string.h> void printlength(char *s, char *t) { unsigned int c = 0; int len = ((strlen(s) - strlen(t)) > c) ? strlen(s): strlen(t); printf (“%d\n”, len); } void main() { char *x = “abc”; char *y = “defgh"; printlength (x,y); } Recall that strlen is defined in string.h as retuming a value of type size_t. which is an unsigned int. The output of the program is =====PAGE 24=====

2017 CS1 · Q.55 · 2 marks [OCR]
The output of executing the following C program is #include <stdio.h> int=total (int -v) { static int count = 0; while(v) { count += vél; v>>= 1; } return count; } void main ()., { statig ime x = 0; int i= 5; for: 4. 08 a") # Ke total(a)- } printé (“Sd\n"”,. x)? =====PAGE 25=====

2017 CS2 · Q.2 · 1 mark [OCR]
Match the following: (P) static char var; (i) Sequence of memory locations to store addresses (Q)m = malloc(10); (ii) A variable located in data section of memory m = NULL; (R) char *ptr[10]; (iii) Request to allocate a CPU register to store data (S) register int varl; (iv) A lost memory which cannot be freed
(A) P = (ii), Q > (iv). R> (). S > iii)
(B) P = (ii), Q—> (@), R— (iv). S — (iii)
(C) P= (ii). Q > (iv). R= Gi). S
(D) P > iii), Q > (iv), R> (@). S > Gi)

2017 CS2 · Q.14 · 1 mark [OCR]
Consider the following function implemented in C: void printxy(int x, int y) { ink. per? x = 0; ptr = &x; y = *per? *ptr = 1; pEANEL(R4),S0” 2 W)it The output of invoking printxy (1,1) is
(A) 0.0
(B) 0.1 (Cc). 1.0
(D) 1.1

2017 CS2 · Q.37 · 2 marks [OCR]
Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x, y. q and r are all unsigned int. while (r >= y) { r= 2 - ye g= iat Ly } Which of the following conditions on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition x == (y*q + my?
(A) (q == r) && (x == 0)
(B) (x > 0) && (r == x) && (y >°0)
(C) (q == 0) && (r == x) && (y > 0) @M) (q == 0) && (y > 0)

2017 CS2 · Q.38 · 2 marks [OCR]
Consider the following C function. int: fun (ant n) af Int shy |} for(ic=: be kk <= ne SEA) 4 for(] = chy -< mye: 7 tS a) print£(TM td Sd” 1,4); } } Time complexity of fun in terms of © notation is
(A) O(nyn)
(B) O(n?)
(C) O(n log n)
(D) O(n? logn) =====PAGE 15=====

2017 CS2 · Q.43 · 2 marks [OCR]
Consider the following snippet of a C program. Assume that swap (&x, &y) exchanges the contents of x and y. int main() { int. array [] = {8; 5S, iy 4, 6, 2h int done = 0; int: a5 while (done == 0) { done = 1; for (i=0; i<=4; i++) { if (array[i] < array[it1]) #4 swap(&array[i], f&array[it+l]); done = 0; } } for (i=5; i>=1; i--) { if (array[i] >/array[i-1]) { swap(&array[{i], &array[i-1l]); done = 0; } print£ (”3d”, arnayP?3]) + } The output of the program is =====PAGE 18=====

2017 CS2 · Q.54 · 2 marks [OCR]
Consider the following C Program. #include<stdio.h> int main() { int m = 10; imi: ny; nie n = ++m; nl = mt++; A=; og a Le n -= nl; printf ("td"; nj? return 0; } The output of the program is

2017 CS2 · Q.55 · 2 marks [OCR]
Consider the following C Program. #include<stdio.h> #include<string.h> int main() { char* c = "GATECSIT2017"; char* p = c; printi("td", (int) strlen (c+2 [p)]—6[p]-1))? return 0; The output of the program is =====PAGE 23=====

2016 · Q.12 · 1 mark
Consider the following C program. void f(int, short); void main() { int i = 100; short s = 12; short *p = &s; _____ ; // call to f() } Which one of the following expressions, when placed in the blank above, will NOT result in a type checking error?
(A) f(s,*s)
(B) i = f(i,s)
(C) f(i,*s)
(D) f(i,*p)

2016 · Q.15 · 1 mark
Consider the following C program. #include<stdio.h> void mystery(int *ptra, int *ptrb) { int *temp; temp = ptrb; ptrb = ptra; ptra = temp; } int main() { int a=2016, b=0, c=4, d=42; mystery(&a, &b); if (a < c) mystery(&c, &a); mystery(&a, &d); printf("%d\n", a); } The output of the program is .

2016 · Q.34 · 2 marks
The following function computes the maximum value contained in an integer array p[] of size n (n >= 1). int max(int *p, int n) { int a=0, b=n-1; while (_____) { if (p[a] <= p[b]) { a = a+1; } else { b = b-1; } } return p[a]; } The missing loop condition is
(A) a != n
(B) b != 0
(C) b > (a + 1)
(D) b != a

> **2016 · Q.35 · 2 marks**
> What will be the output of the following C program? void count(int n){ static int d=1; printf("%d ", n); printf("%d ", d); d++; if(n>1) count(n-1); printf("%d ", d); } void main(){ count(3); }
> (A) 3 1 2 2 1 3 4 4 4
> (B) 3 1 2 1 1 1 2 2 2
> (C) 3 1 2 2 1 3 4
> (D) 3 1 2 1 1 1 2

B. New practice questions with solutions

1. (1 mark · MCQ) What does the program print?

```
int a[5] = {10, 20, 30, 40, 50};
int *p = a;
p += 2;
printf("%d %d %ld", *p, p[-1], (long)(&a[4] - p));
```

A. 30 20 2 (B) 30 20 3 (C) 20 30 2 (D) 30 30 2

> **(A).** p points to a[2] = 30, p[-1] = a[1] = 20, and &a[4] − p = 2 elements.

2. (2 marks · MCQ) What does the program print?

```
int c = 0;
int g(int n) {
    c++;
    if (n <= 1) return n;
    return g(n-1) + g(n-2);
}
int main() {
    int v = g(5);
    printf("%d %d", v, c);
}
```

A. 5 9 (B) 5 15 (C) 8 15 (D) 5 25

> **(B).** g(5) is the 5th Fibonacci number, 5. The number of calls satisfies calls(n) = 1 + calls(n−1) + calls(n−2) with calls(0) = calls(1) = 1, giving 1, 1, 3, 5, 9, 15.

Trees, BST, heaps

BST/heap properties, traversals, heap construction, tree height and node counts. · 29 questions, 44 marks over 15 papers (≈2.9 marks/paper)

A. Past-paper questions (29)

2026 CS1 · Q.23 · 1 mark

Let n be an odd number greater than 100. Consider a binary minheap with n elements stored in an array P whose index starts from 1. Which of the following indices of P do/does NOT correspond to any leaf node of the minheap?
(A) n+1 2
(B) n−1 2
(C) n−3 2
(D) n

2026 CS1 · Q.33 · 1 mark

The height of a binary tree is the number of edges in the longest path from the root to a leaf in the tree. The maximum possible height of a full binary tree with 23 nodes is _____. (answer in integer)

2026 CS1 · Q.40 · 2 marks

Let P be the set of all integers from 1 to 15. Consider any order of insertion of the elements of P into a binary search tree that creates a complete binary tree. Which one of the following elements can NEVER be the third element that is inserted?
(A) 4
(B) 2
(C) 10
(D) 5

2026 CS1 · Q.62 · 2 marks

The following sequence corresponds to the preorder traversal of a binary search tree T: 50, 25, 13, 40, 30, 47, 75, 60, 70, 80, 77 The position of the element 60 in the postorder traversal of T is _____. (answer in integer) Note: The position begins with 1.

2026 CS2 · Q.12 · 1 mark

The set T represents various traversals over binary tree. The set S represents the order of visiting nodes during a traversal. T S I: Inorder L: left subtree, node, right subtree II: Preorder M: node, left subtree, right subtree III: Postorder N: left subtree, right subtree, node Which one of the following is the correct match from T to S ?
(A) I – L, II – M, III – N
(B) I – M, II – L, III – N
(C) I – N, II – M, III – L
(D) I – L, II – N, III – M

2026 CS2 · Q.49 · 2 marks
Consider a binary search tree (BST) with n leaf nodes (n > 0). Given any node V, the key present in the node is denoted as Val(V). All the keys present in the given BST are distinct. The keys belong to the set of real numbers. For a node V, let Suc(V) denote the node that is its inorder successor. If a node V does not have an inorder successor, then Suc(V) is NULL. As there are no duplicates, if Suc(V) is not NULL, then Val(V) < Val(Suc(V)). Corresponding to every leaf node Li that has a non-NULL Suc(Li), a new key ki with the following property is to be inserted into the BST. Val(Li) < ki < Val(Suc(Li)) Let K represent the list of all such new keys to be inserted into the BST. Which of the following statements is/are true?
(A) K cannot have any duplicates
(B) K will have at least one element
(C) After inserting all keys from K, the height of the BST can increase at most by one
(D) Number of nodes in the BST will double after inserting all keys from K

2025 CS1 · Q.26 · 1 mark
Which of the following statement(s) is/are TRUE for any binary search tree (BST) having n distinct integers?
(A) The maximum length of a path from the root node to any other node is (n − 1).
(B) An inorder traversal will always produce a sorted sequence of elements.
(C) Finding an element takes O(log 2 n) time in the worst case.
(D) Every BST is also a Min-Heap.

2025 CS1 · Q.35 · 1 mark
The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node. Suppose a Min-Heap T stores 32 keys. The height of T is _____. (Answer in integer)

2025 CS2 · Q.13 · 1 mark
Consider a binary tree T in which every node has either zero or two children. Let n > 0 be the number of nodes in T. Which ONE of the following is the number of nodes in T that have exactly two children?
(A) n−2 2
(B) n−1 2
(C) n 2
(D) n+1 2

2025 CS2 · Q.35 · 1 mark
Suppose the values 10, −4, 15, 30, 20, 5, 60, 19 are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. The number of edges in the path from the node containing 19 to the root node of T is _____. (Answer in integer)

2025 CS2 · Q.38 · 2 marks
A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures: P: Unsorted doubly linked list with pointers to the head node and tail node of the list. Q: Min-heap implemented using an array. R: Binary Search Tree. Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?
(A) P: Θ(1), Q: Θ(n), R: Θ(n)
(B) P: Θ(1), Q: Θ(n log n), R: Θ(n)
(C) P: Θ(n), Q: Θ(n log n), R: Θ(n2)
(D) P: Θ(1), Q: Θ(n), R: Θ(n log n)

2024 CS1 · Q.41 · 2 marks
An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?
(A) 82, 90, 101
(B) 82, 11, 93
(C) 131, 11, 93
(D) 131, 111, 90

2024 CS1 · Q.43 · 2 marks
Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is
(A) 53
(B) 52
(C) 27
(D) 1

2024 CS2 · Q.39 · 2 marks
You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T, and you do not have access to T. Which one of the following statements is TRUE?
(A) Inorder traversal of T can be determined from V
(B) Root node of T can be determined from V
(C) Preorder traversal of T can be determined from V
(D) Postorder traversal of T can be determined from V

2023 · Q.12 · 1 mark
Which one of the following sequences when stored in an array at locations A[1], . . . , A[10] forms a max-heap?
(A) 23, 17, 10, 6, 13, 14, 1, 5, 7, 12
(B) 23, 17, 14, 7, 13, 10, 1, 5, 6, 12
(C) 23, 17, 14, 6, 13, 10, 1, 5, 7, 15
(D) 23, 14, 17, 1, 10, 13, 16, 12, 7, 5 CS

2023 · Q.46 · 2 marks

Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation Extract-Max
(A) extracts and deletes the maximum element from A. The operation Insert(A,key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?
(A) Both Extract-Max
(A) and Insert(A,key) run in O(1).
(B) Both Extract-Max
(A) and Insert(A,key) run in O(log(n)).
(C) Extract-Max
(A) runs in O(1) whereas Insert(A,key) runs in O(n).
(D) Extract-Max
(A) runs in O(1) whereas Insert(A,key) runs in O(log(n)). CS

2023 · Q.47 · 2 marks

Consider the C function foo and the binary tree shown. typedef struct node { int val; 10 struct node *left, *right; } node; 5 11 int foo(node *p) { int retval; 3 8 13 if (p == NULL) return 0; else { retval = p->val + foo(p->left) + foo(p->right); printf("%d ", retval); return retval; } } When foo is called with a pointer to the root node of the given binary tree, what will it print?
(A) 3 8 5 13 11 10
(B) 3 5 8 10 11 13
(C) 3 8 16 13 24 50
(D) 3 16 8 50 24 13 CS

2022 · Q.28 · 1 mark

Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index_____.

2021 · Q.10 · 1 mark [OCR]

A binary search tree T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T?
(A) | @(nlogn)
(B) | O(n)
(C) | O(log n)
(D) | 9) CS, Set-1 - Copyright © GATE 2021

2020 · Q.5 · 1 mark [OCR]

The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19. Which one of the following is the postorder traversal of the tree?
(A) 10, 11, 12, 15, 16, 18, 19, 20 (8) 11, 12, 10, 16, 19, 18, 20, 15 © 20, 19, 18, 16, 15, 12, 11, 10 (0) 19, 16, 18, 20, 11, 12, 10, 15

2020 · Q.6 · 1 mark [OCR]
What is the worst case time complexity of inserting n? elements into an AVL-tree with n elements initially? (a) @(n*) 6) 0(n?) © Q(n?log n) ©) O(n3)

2020 · Q.41 · 2 marks [OCR]
In a balanced binary search tree with n elements, what is the worst case time complexity of reporting all elements in range [a,b]? Assume that the number of reported elements is k. (a) O(log n) (8) O(logn + k) () O(klogn) (0) O(nlog k)

2020 · Q.47 · 2 marks [OCR]
Consider the array representation of a binary min-heap containing 1023 elements. The minimum number of comparisons required to find the maximum in the heap is

2019 · Q.40 · 2 marks [OCR]
Consider the following statements: I. The smallest element in a max-heap is always at a leaf node IL The second largest element in a max-heap is always a child of the root node If. A max-heap can be constructed from a binary search tree in O(n) time IV. _ A binary search tree can be constructed from a max-heap in @(7) time Which of the above statements are TRUE? (A), and I
(B) L and Iv
(C) L and IV
(D) I, Mand IV

2018 · Q.20 · 1 mark
The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

2018 · Q.46 · 2 marks
The number of possible min-heaps containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once is _____. CS 16/20 GATE 2018

2017 CS1 · Q.6 · 1 mark [OCR]
Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are: Note: The height of a tree with a single node is 0.
(A) 4 and 15 respectively
(B) 3 and 14 respectively
(C) 4 and 14 respectively
(D) 3 and 15 respectively

2017 CS2 · Q.36 · 2 marks [OCR]
The pre-order traversal of a binary search tree is given by 12,8,6,2,7,9, 10,16, 15,19, 17,20. Then the post-order traversal of this tree is:
(A) 2,6,7,8,9, 10, 12,15, 16, 17,19, 20 (B):2,7;6,10;9;58; 15,47, 20,19, 16,42
(G)'7;2;6;8;9,10, 20,17; 19,15; 16,42 (D)'7;6;2,10,9; 8,:15;16, 17, 20:19 12

2016 · Q.37 · 2 marks

An operator delete(i) for a binary heap data structure is to be designed to delete the item in the i-th node. Assume that the heap is implemented in an array and i refers to the i-th index of the array. If the heap tree has depth d (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?

(A) O(1)

(B) O(d) but not O(1)

(C) O(2d) but not O(d)

(D) O(d 2d) but not O(2d)

B. New practice questions with solutions

1. (1 mark · MCQ) The keys 10, 20, 15, 30, 25 are inserted one at a time into an initially empty max-heap, using sift-up on each insertion. The final array is

A. 30 25 15 10 20
B. 30 20 25 10 15
C. 30 25 20 10 15
D. 30 20 15 10 25

(A). After 10, 20, 15 the heap is [20, 10, 15]. Inserting 30 gives [30, 20, 15, 10]. Inserting 25 at index 4 swaps it with its parent 20, giving [30, 25, 15, 10, 20].

2. (2 marks · MCQ) A BST is built by inserting 50, 30, 70, 20, 40, 60, 80, 35, 45 in that order. Node 30 is then deleted and replaced by its inorder successor. The preorder traversal of the result is

A. 50 35 20 40 45 70 60 80
B. 50 40 20 35 45 70 60 80
C. 50 35 20 45 40 70 60 80
D. 50 20 40 35 45 70 60 80

(A). The inorder successor of 30 is the minimum of its right subtree, which is 35. Node 30 takes the value 35 and the old leaf 35 is removed. The tree is 50 (35 (20, 40 (–, 45)), 70 (60, 80)).

Stacks, queues, linked lists

Stack/queue operations and sequences, linked-list code. · 16 questions, 24 marks over 15 papers (≈1.6 marks/paper)

A. Past-paper questions (16)

2026 CS1 · Q.39 · 2 marks

Consider the following code snippet in C language that computes the number of nodes in a non-empty singly linked list pointed to by the pointer variable head. struct node{ int elt; struct node *next; }; int getListSize (struct node *head) { if(E1) return 1; return E2; } Which one of the following options gives the correct replacements for the expressions E1 and E2?

(A) E1: head == NULL E2: 1 + getListSize(head)

(B) E1: head->next == NULL E2: 1 + getListSize(head->next)

(C) E1: head == NULL E2: 1 + getListSize(head->next)

(D) E1: head->next == NULL E2: 1 + getListSize(head)

2026 CS2 · Q.50 · 2 marks

Consider a stack S and a queue Q. Both of them are initially empty and have the capacity to store ten elements each. The elements 1, 2, 3, 4, and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeueing all elements. The output obtained by following this process is 4 3 1 2 5 . Given the output, the objective is to predict whether an element was assigned to S or Q. Which of the following options is/are possible valid assignment(s) of the elements? Note: In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q.

(A) 1S, 2Q, 3S, 4S, 5Q

(B) 1Q, 2Q, 3S, 4S, 5Q

(C) 1Q, 2Q, 3Q, 4S, 5S

(D) 1S, 2S, 3S, 4Q, 5Q

2025 CS1 · Q.62 · 2 marks

Let LIST be a datatype for an implementation of linked list defined as follows: typedef struct list { int data; struct list *next; } LIST; Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes. Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____ . (Answer in integer) int find (int query, LIST *list) { while (list != NULL){ if(list->data == query) return 1; list = list->next; } return 0; } int main () { ... ... ptr1=L1; ptr2=L2; while (ptr1->next != NULL) { query = ptr1->next->data; if (find (query, L2)) ptr1->next = ptr1->next->next; else ptr1 = ptr1->next; } ... ... return 0; }

2025 CS2 · Q.45 · 2 marks

Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an O(1) time MIN operation that returns a pointer to the record with smallest key present in the stack 1) without deleting the corresponding record, and 2) without increasing the complexities of the standard stack operations. Which one or more of the following approach(es) can achieve it?
(A) Keep with every record in the stack, a pointer to the record with the smallest key below it.
(B) Keep a pointer to the record with the smallest key in the stack.
(C) Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
(D) Keep a Min-Heap in which the key values of the records in the stack are maintained.

2024 CS2 · Q.48 · 2 marks

Let S1 and S2 be two stacks. S1 has capacity of 4 elements. S2 has capacity of 2 elements. S1 already has 4 elements: 100, 200, 300, and 400, whereas S2 is empty, as shown below. 400 (Top) 300 200 Stack S2 100 Stack S1 Only the following three operations are available: PushToS2: Pop the top element from S1 and push it on S2. PushToS1: Pop the top element from S2 and push it on S1. GenerateOutput: Pop the top element from S1 and output it to the user. Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack. Which of the following output sequences can be generated by using the above operations?
(A) 100, 200, 400, 300
(B) 200, 300, 400, 100
(C) 400, 200, 100, 300
(D) 300, 200, 400, 100

2023 · Q.13 · 1 mark

Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list. Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?
(A) SLLdel is O(1) and DLLdel is O(n)
(B) Both SLLdel and DLLdel are O(log(n))
(C) Both SLLdel and DLLdel are O(1)
(D) SLLdel is O(n) and DLLdel is O(1) CS

2023 · Q.59 · 2 marks

Consider a sequence a of elements a0 = 1, a1 = 5, a2 = 7, a3 = 8, a4 = 9, and a5 = 2. The following operations are performed on a stack S and a queue Q, both of which are initially empty. I: push the elements of a from a0 to a5 in that order into S. II: enqueue the elements of a from a0 to a5 in that order into Q. III: pop an element from S. IV: dequeue an element from Q. V: pop an element from S. VI: dequeue an element from Q. VII: dequeue an element from Q and push the same element into S. VIII: Repeat operation VII three times. IX: pop an element from S. X: pop an element from S. The top element of S after executing the above operations is . CS

2022 · Q.15 · 1 mark
Consider the problem of reversing a singly linked list. To take an example, given the linked list below, the reversed linked list should look like Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in O (1) space?
(A) The best algorithm for the problem takes (n) time in the worst case.
(B) The best algorithm for the problem takes (n log n) time in the worst case.
(C) The best algorithm for the problem takes (n2) time in the worst case.
(D) It is not possible to reverse a singly linked list in O (1) space.

2022 · Q.62 · 2 marks
Consider the queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue(Q,element) and Dequeue(Q). The minimum number of Enqueue operations on Q1 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage is_____.

2021 · Q.21 · 1 mark [OCR]
Consider the following sequence of operations on an empty stack. push(54); push(52); pop(); push(55); push(62); s = pop(); Consider the following sequence of operations on an empty queue. enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32);q = dequeue(): The value of s + q is

2020 · Q.16 · 1 mark [OCR]
What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?
(A) O(n) (8) O(nlogn) © O(n?)
(D) 0(1)

2018 · Q.3 · 1 mark
A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail. head tail Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?
(A) θ(1), θ(1)
(B) θ(1), θ(n)
(C) θ(n), θ(1)
(D) θ(n), θ(n) CS 1/20 GATE 2018

2017 CS1 · Q.8 · 1 mark [OCR]
Consider the C code fragment given below. typedef struct node { int data; node* next; } node; void join(node* m, node* n) { node* p = n; while(p->next != NULL) { Pp = p->next; } p->next = m; } Assuming that m and n point to valid NULL-terminated linked lists. invocation of j oin will
(A) append list m to the end of list n for all inputs.
(B) either cause a null pointer dereference or append list m to the end of list n.
(C) cause a null pointer dereference for all inputs.
(D) append list n to the end of list m for all inputs.

2017 CS2 · Q.13 · 1 mark [OCR]
A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers FRONT and REAR pointing to the front node and the rear node of the queue. respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in O(1) time? I. Next pointer of front node points to the rear node. Il. Next pointer of rear node points to the front node.
(A) Tonly
(B) II only
(C) Both I and II
(D) Neither I nor II =====PAGE 6=====

2016 · Q.10 · 1 mark
A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?
(A) Both operations can be performed in O(1) time
(B) At most one operation can be performed in O(1) time but the worst case time for the other operation will be Ω(n)
(C) The worst case time complexity for both operations will be Ω(n)
(D) Worst case time complexity for both operations will be Ω(log n)

2016 · Q.41 · 2 marks
Let Q denote a queue containing sixteen numbers and S be an empty stack. Head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly Top(S) returns the element at the top of S without removing it from S. Consider the algorithm given below. while Q is not Empty do if S is Empty OR Top(S) ≤ Head(Q) then x := Dequeue(Q); Push(S, x); else x := Pop(S); Enqueue(Q, x); end end The maximum possible number of iterations of the while loop in the algorithm is .

B. New practice questions with solutions

1. (1 mark · MCQ) An empty stack undergoes these operations in order: push 1, push 2, push 3, pop, push 4, pop, pop, push 5, pop. The values popped, in order, are

A. 3 4 2 5 (B) 3 4 1 5 (C) 3 2 4 5 (D) 3 4 2 1

(A). The pops return 3, then 4, then 2, then 5. The value 1 stays in the stack.

2. (2 marks · NAT) A singly linked list holds 1 → 2 → 3 → 4 → 5. The function below is called with the head. The data in the third node of the returned list is _____.

```
struct node { int d; struct node *next; };
struct node *g(struct node *h) {
    struct node *a = h, *b;
    if (!a || !a->next) return a;
    b = a->next;
    a->next = g(b->next);
    b->next = a;
    return b;
}
```

> **4.** The function swaps adjacent pairs, so the result is 2 → 1 → 4 → 3 → 5.